

MOBILE CHARGER

-A technical story

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1.Main Objective:

- To design and build a functional 5V mobile charger using a transformer, bridge rectifier, and voltage regulator.
- To understand the basic stages of converting 230V AC household power into stable 5V DC.
- To identify real-world issues like heat generation and slow charging through testing, and explore engineering solutions to improve the charger's performance.

2.Final Components:

Materials	Uses
Step down transformer	Converts 230v AC To 12v AC
Perfboard	To physically mount and solder components
Full wave bridge rectifier(4-IN4007 Diodes)	Converts AC voltage to pulsating DC Voltage
Capacitors(1000uF, 0.1uF)	Smooths pulsating DC to steady DC and 1uF filters high frequency noise
LM2596 Buck Converter module	Step down DC-DC converter
USB	Used to connect and charge
Connecting wires	Connect all components

3.Our Journey

Phase 1: Initial Design & Breadboard Testing

➤ Components Selected:

We began by assembling the following components:

Step-down Transformer: Converts 230V AC to 12V AC.

Perfboard: To physically mount and solder the components.

Full-Wave Bridge Rectifier (4x 1N4007 Diodes): Converts AC voltage to pulsating DC voltage.

Capacitors (1000 μ F, 1 μ F): Smooths the pulsating DC to steady DC; the 1 μ F filters high-frequency noise.

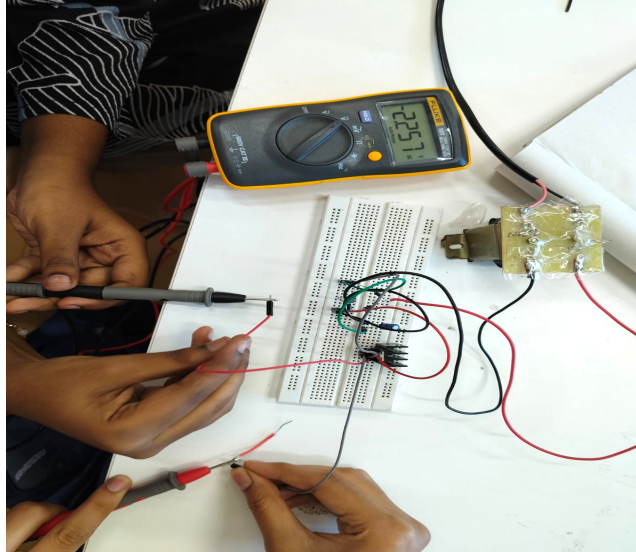
7805 Voltage Regulator with heatsink(Initially we used this): Regulates 8-17V to a stable 5V output.

USB Port: Used to connect and charge the phone.

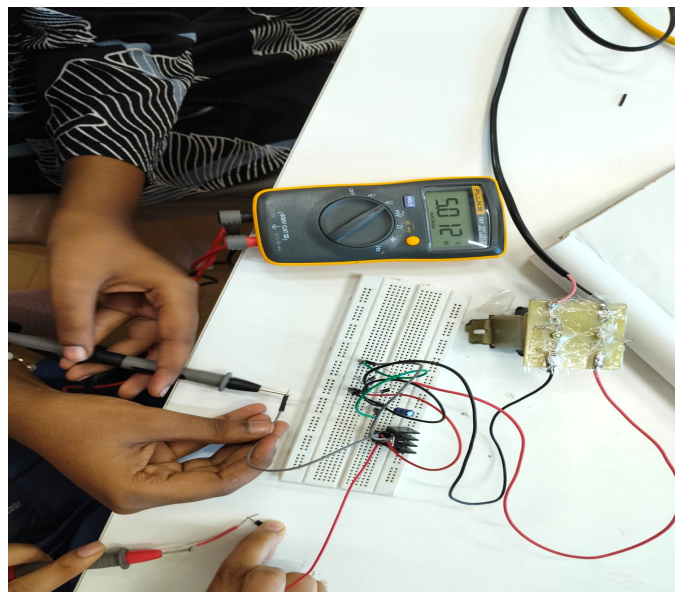
Before soldering onto the perfboard, we set up the circuit on a breadboard to verify connections and avoid multiple soldering/desoldering cycles. We also soldered the output wires onto the transformer, as it came without them.

Problem 1: Unexpected Voltage Reading

Issue: During our first breadboard test, the multimeter displayed an unexpected 2V instead of the expected 5V.



Solution: We dismantled and reassembled the circuit. The issue was diagnosed as a loose or improper connection. Upon reassembly, the circuit correctly outputted 4.8V to 5.1V.



➤ **Research and Component Optimization:**

We researched ways to make the circuit more efficient and address common charger problems (overheating, slow charging, voltage fluctuation). Potential solutions included adding a heatsink, capacitors, resistors. (Note: A heatsink was already attached to the voltage regulator).

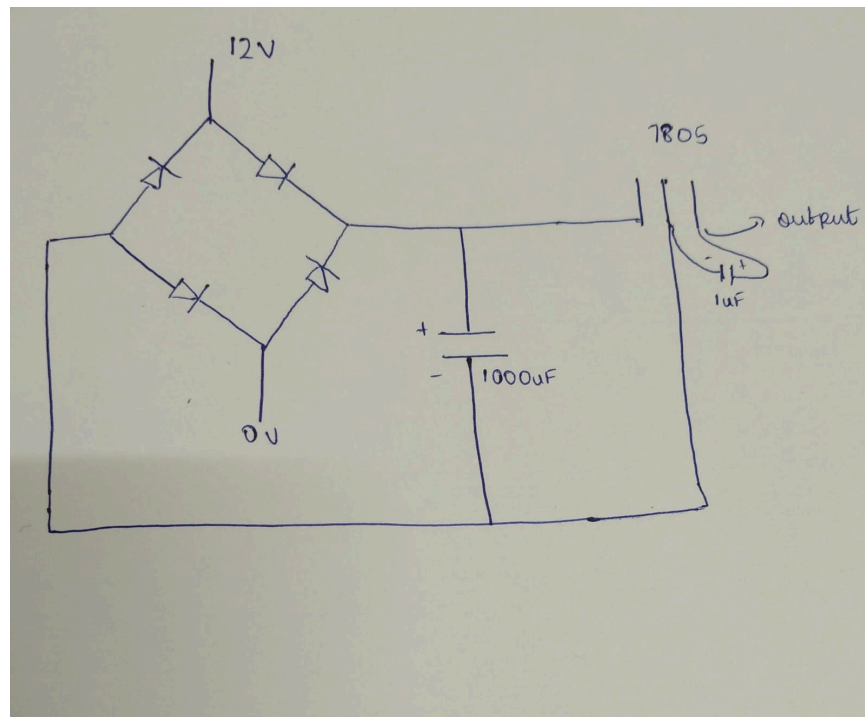
➤ Testing an Alternative Configuration:

Keeping the rectifier unchanged, we tried another configuration: 1000 μ F and 1 μ F capacitors in parallel, a 100 μ F capacitor connected to 7805 pins 2 and 3, and two resistors (47k Ω and 56k Ω) in series between pins 2 and 3.

Result: The circuit worked i.e we got 5v with multimeter, but we later found that the 100 μ F capacitor and the two resistors were redundant.

Why: The 1000 μ F capacitor is already ten times larger and handles all the heavy smoothing, making the 100 μ F unnecessary clutter. The resistors are critically redundant because they are meant for adjustable regulators where specific resistor values determine output voltage. The 7805 is a fixed 5V regulator; adding these resistors would drop the voltage below 5V.

Phase 2: Perfboard Assembly & Soldering Challenges



➤ **Layout Planning:**

We planned the layout to save space. We placed the 4 diodes first, twisting the legs of adjacent diodes to form the full-wave bridge rectifier, and soldered them to secure the joints. We placed the 1000 μ F capacitor, ensuring its negative leg (white stripe) was correctly positioned. The 1 μ F capacitor was connected to Pin 3 (output) and Pin 2.

Critical Note: We kept all other components away from the 7805 and heatsink to prevent melting or burning from the heat.

Problem 2: Soldering Gun Failure

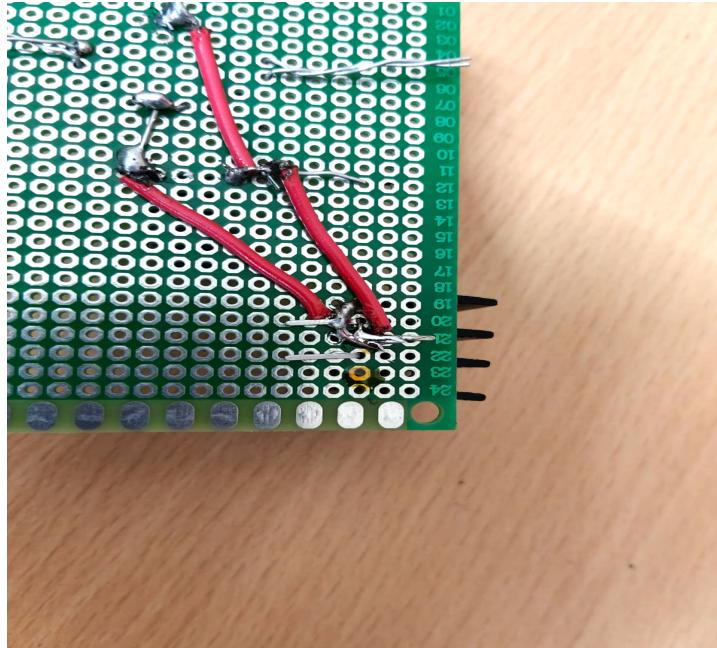
Issue: Our soldering gun was not melting the metal properly, forming metal balls that fell off. We lacked flux.

Attempted Fixes: We asked peers and AI for advice, tried filing the tip, and rubbed it on hard surfaces to remove rust/impurities. It still didn't work.

Solution: With special thanks to Lasya, we acquired a new soldering gun, proper components, and flux. We resumed soldering.

Problem 3: Solder Bridge on the 7805

Issue: While soldering the pins of the voltage regulator, one pin accidentally touched the adjacent metal pin of 7805, creating a solder bridge. While trying to desolder and move the pin, the melted metal fell between the pins, making the connection even worse.

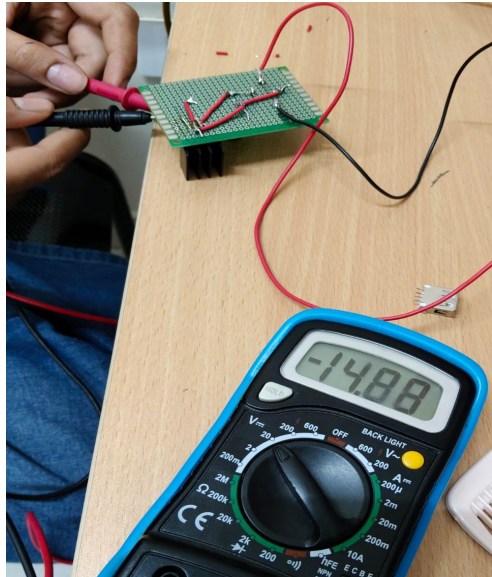


But still we tried to melt it again and then somehow removed the solder blobs.

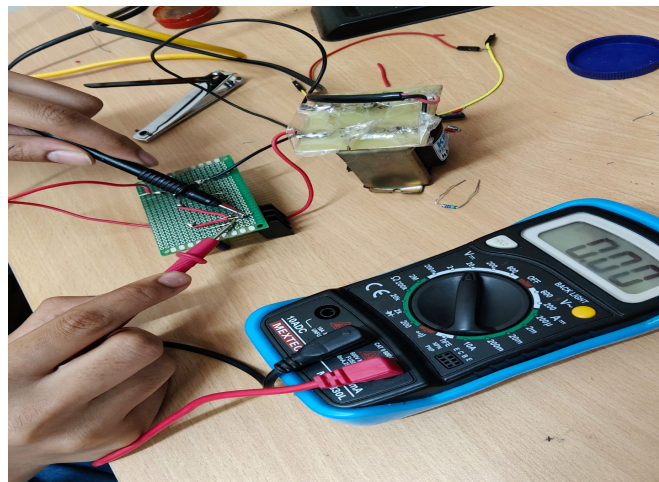
Additional Note: And another thing is that we connected pins of 7805 (pin 3) directly to capacitor leg and because of soldering at very high temperature I guess we damaged it as the high temperature would boil the liquid inside.

Phase 3: Testing Failure & Short Circuit

Problem 4: The 14V Reading and Short Circuit



Issue: During testing, the multimeter displayed 14V instead of 5V. Before we could properly diagnose it, we randomly moved the multimeter probes. The reading suddenly dropped to 0V and stayed there, no matter where we connected the probes.

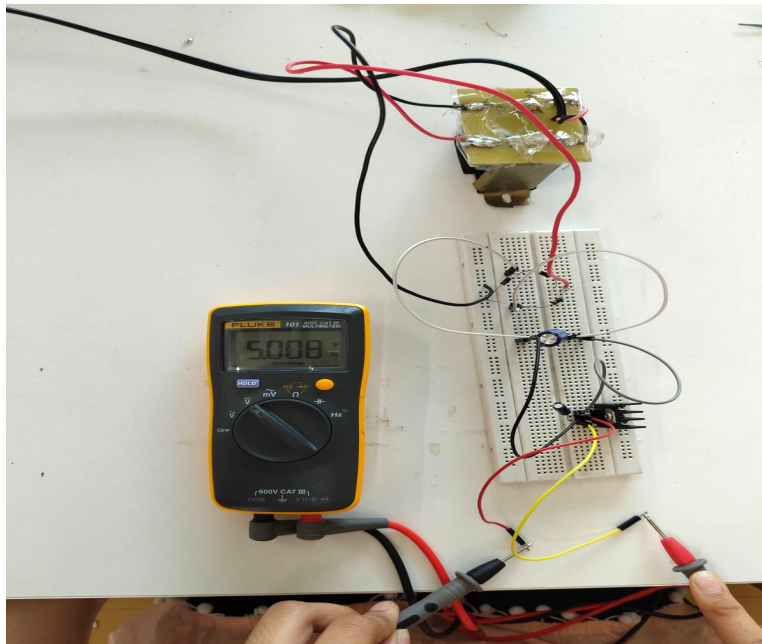


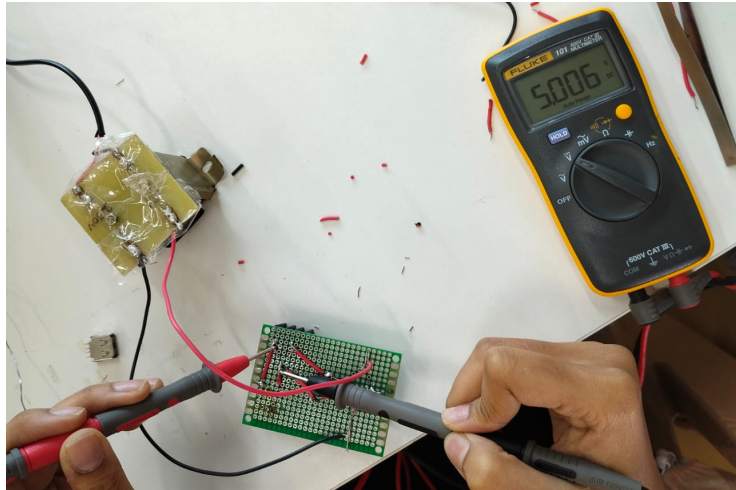
Diagnosis: We realized we had accidentally connected the capacitor pins directly to each other, creating a short circuit. This damaged the 7805 voltage regulator.

Consequence: We didn't have an extra 7805, and the labs were closed. We couldn't do anything until Monday, forcing a break in soldering.

Phase 4: Circuit Reconstruction & Verification

We desoldered the damaged 7805 and 1 μ F capacitor and removed excess solder. We issued new components and rebuilt the circuit on a breadboard (using 2 capacitors instead of 3 as initially planned). It successfully outputted 5V, so we moved to soldering the final components.





Working Principle (The Final Stages):

Transformer (230V AC $\rightarrow$ 12V AC): Steps down deadly 230V to safe 12V and physically isolates the phone from the wall.

Bridge Rectifier (12V AC $\rightarrow$ Pulsing DC): Four 1N4007 diodes form a diamond. In both half-cycles, current is forced to exit from the Top Node (+). Output is pulsing DC (0V to 17V, 100 times a second).

1000 μ F Capacitor (Pulsing DC $\rightarrow$ Steady 17V DC): Acts like a water tank. Charges at 17V, releases energy at 0V. Provides a steady input so the regulator doesn't overheat.

7805 Regulator (17V DC $\rightarrow$ 5V DC): Constantly measures output. If it drops below 5V, it opens the transistor; if it rises, it closes. It burns off the extra 12V as heat, ensuring the phone gets exactly 5V.

1 μ F Capacitor (Clean 5V $\rightarrow$ Cleaner 5V): The 7805 creates high-frequency spikes. The 1 μ F acts as a sponge, catching them and sending them to Ground. Without it, the 5V is dirty, and the touchscreen may act weird.

USB Port (The Handshake): We shorted the D+ and D- pins together.

This signals to the phone, "I am a dedicated wall charger, give me full power," doubling charging speed from 0.5A to 1A.

Result: When we checked the whole setup with an app named Ampereflow We got the power fluctuating a lot, like it started at 2.7watts and then reduced to 0.5watts and started fluctuating between 0.5watts to 1 watts which is very less as compared to modern chargers which deliver upto 15watts. This is because Linear regulators burn the excess voltage as heat.

Problem: While the 7805 worked, it was fundamentally inefficient for fast charging. It dropped 17V to 5V by burning 12V as heat. At 1A, it generated 12 Watts of heat and would thermal throttle or shut down.

$$\text{Heat} = \text{voltage drop} * \text{current}$$

$$\text{Power} = V * I$$

$$2.7\text{watts} = 5\text{v} * I$$

$$I = 0.54\text{amp}$$

$$\text{Voltage drop} = 17 - 5 = 12\text{v}$$

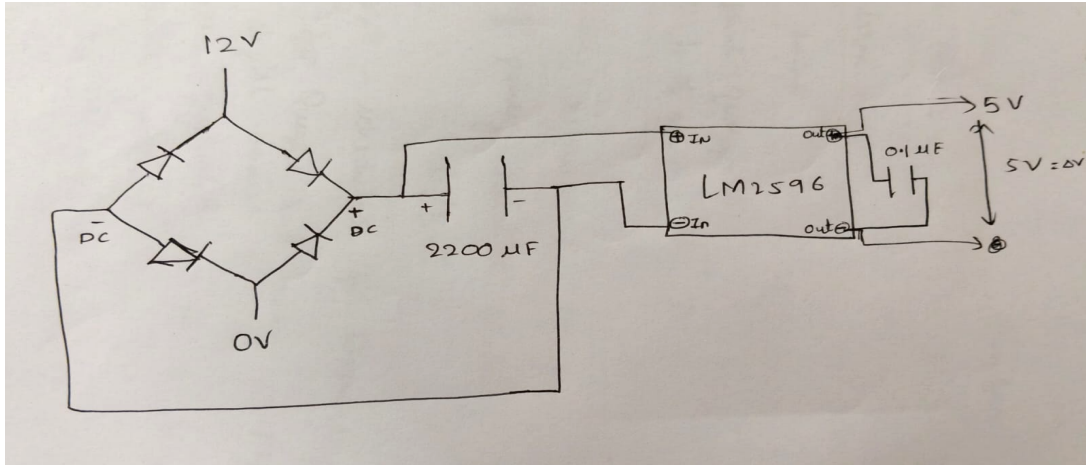
$$\text{Heat} = 12 * 0.54 = 6.5 \text{ watts heat}$$

Therefore 6.5 watt heat load on a tiny heatsink will make the 7805 get heated and shut down for protection.

Why We Upgraded:

To achieve a stable 5V/3A (15W) output without extreme heat, we replaced the linear 7805 with an LM2596 Buck Converter module. The LM2596 is a switching regulator (~85% efficient), meaning it generates minimal heat.

Phase 5: Upgrade to LM2596 Buck Converter



New Components Added:

LM2596 Module: The core switching regulator.

2200 μ F Capacitor: Used for input smoothing to handle the 12V 2A transformer's output.

0.1 μ F Capacitor: Added to the output to filter high-frequency noise.

Makeshift Heatsink (Coin/Aluminium foil): We placed a coin directly on the LM2596 chip to help dissipate heat during testing.

(Note: We did not use any external Schottky diodes or a professional heatsink for this setup).

Testing with breadboard first:

After wiring the transformer, bridge rectifier, and 2200 μ F capacitor to the LM2596 module, we calibrated the potentiometer to output exactly 5.0V.

→ But even with LM2596 we are getting nearly 3-3.5 watts and then it is fluctuating a lot .

→ So we found it is just because of using jumper wires and breadboard

which has very high resistance which is not allowing much current to flow.

→ And then we removed 0.1uF capacitor at the output but it resulted in electrical noise at the beginning. So we connected it again.

Phase 6: Performance Measurement & Final Observations

To verify the final circuit after soldering every component, we measured the following parameters under load:

1. Input Voltage (Transformer Output)

Measurement Point: Across the two output wires of the 12V transformer.

Expected Reading: ~12V AC (no load).

Our Reading: ~13.6V AC. This is normal for a lightly loaded transformer.

2. Rectified Input Voltage (Capacitor)

Measurement Point: Across the 2200μF capacitor (Input to the LM2596).

Expected Reading: ~17V to 19V DC (unloaded).

[Our Reading when connected to breadboard: ~18V DC unloaded, dropping to ~5.7V under heavy load. This severe voltage sag indicated the transformer was maxed out and the LM2596 was struggling.

Solution: We constructed the whole circuit again using normal wires (thick with less resistance) and used coin as heat sink to reduce voltage sagging.]

3. Output Voltage (LM2596)

Measurement Point: Across the USB port's +5V and GND pins.

Reading: 5.06V DC.

4. Output Current & Wattage

Measurement Tool: AmpereFlow App (Phone-side measurement).

Observation: The charger delivered an average of 5 to 6 Watts (5V at ~1A to 1.2A). The wattage fluctuated as the phone's battery level changed and the LM2596 thermally throttled.

We even got 7watts after waiting for a minute, it means that phone is pulling 1.4Amp current

Why it fluctuated: The 12V 2A transformer (24W max) was pushed near its limit, causing voltage sag. The coin heatsink was insufficient for continuous 3A charging, causing the LM2596 to enter thermal throttling mode.

[Click here to see the readings](#)

5. Thermal Check

Observation: The coin placed on the LM2596 sometimes becoming hot during operation. The inductor on the module began to vibrate (whine) under heavy load, indicating stress. However, the circuit did not shut down completely and continued to deliver a steady 5V to the phone.

But after 1 minute diodes and perfboard started heating.

Since at last we are getting 7watts power

Heat at each diode= $v \cdot i$

Voltage drop= $0.7v$

Heat= $0.7 \cdot 1.4 = 0.98 \text{ watts}$

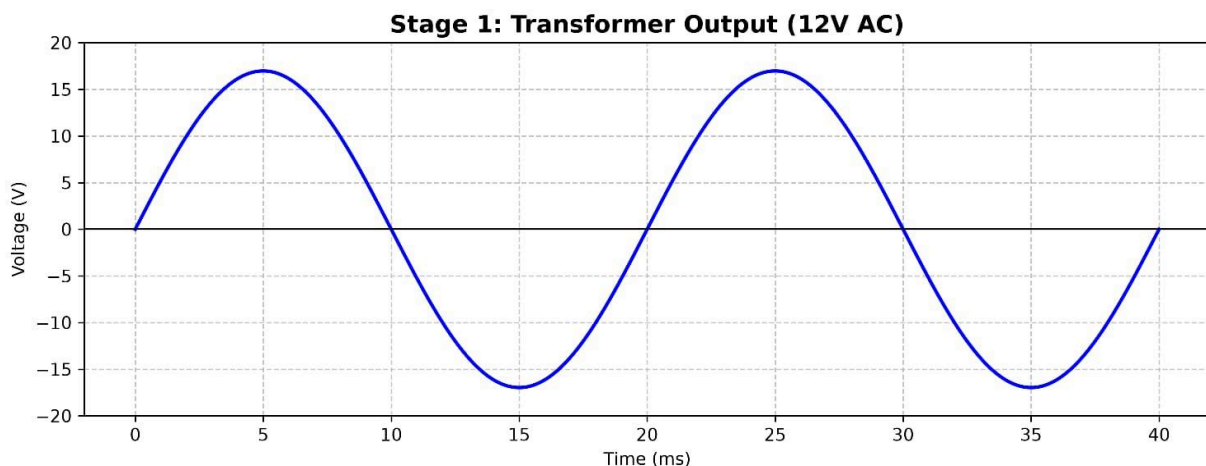
Because current passes through 2 diodes at a time, total heat=2 watts
1N4007 diodes are rated for 1amp. But we are overloading by 40%, they are being pushed beyond limit(but safe for a very short time).

Working principle (BTS):

1. Transformer: 230V AC $\rightarrow$ 12V AC

image

· Steps down deadly 230V to safe 12V and physically isolates your phone from the wall (no direct wire connection).

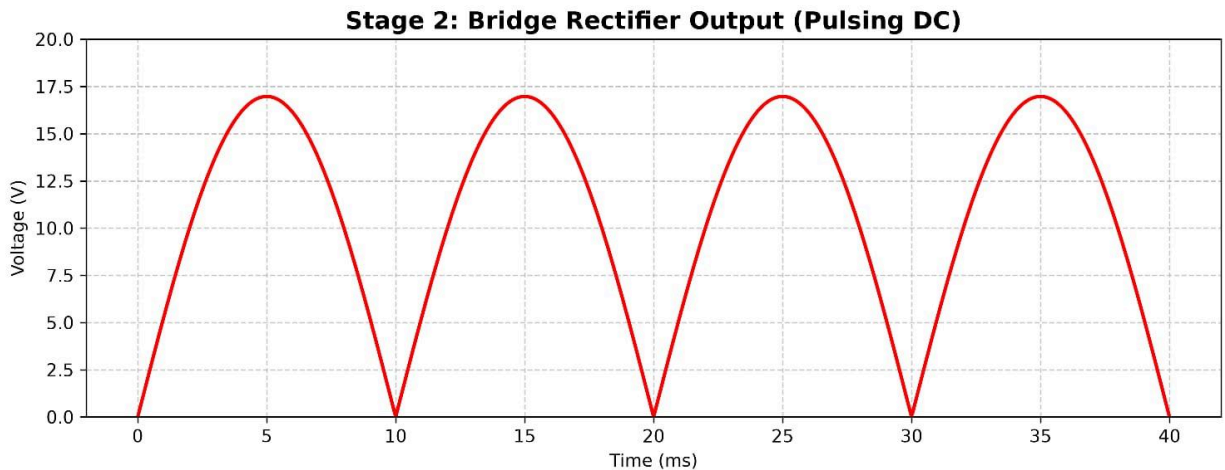


2. Bridge Rectifier: 12V AC $\rightarrow$ Pulsing DC

· Four 1N4007 diodes form a diamond. A diode is a one-way valve—current flows only from Anode (triangle) to Cathode (silver stripe).
· In the positive half-cycle, current flows through the top-left and bottom-right diodes. In the negative half-cycle, it flows through the top-right

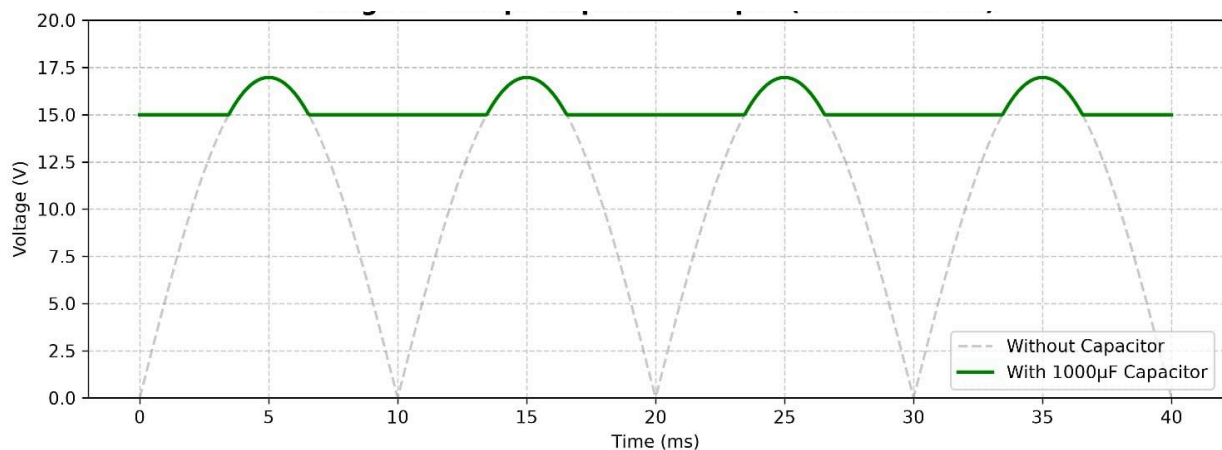
and bottom-left diodes.

- In both cases, current is forced to exit from the Top Node (+).
- The output is pulsing DC (0V to 17V, 100 times a second).
- Forces AC into one direction so the phone battery (which only works with DC) can use it.



3. 2200 μ F Capacitor: Pulsing DC $\rightarrow$ Steady 17V DC

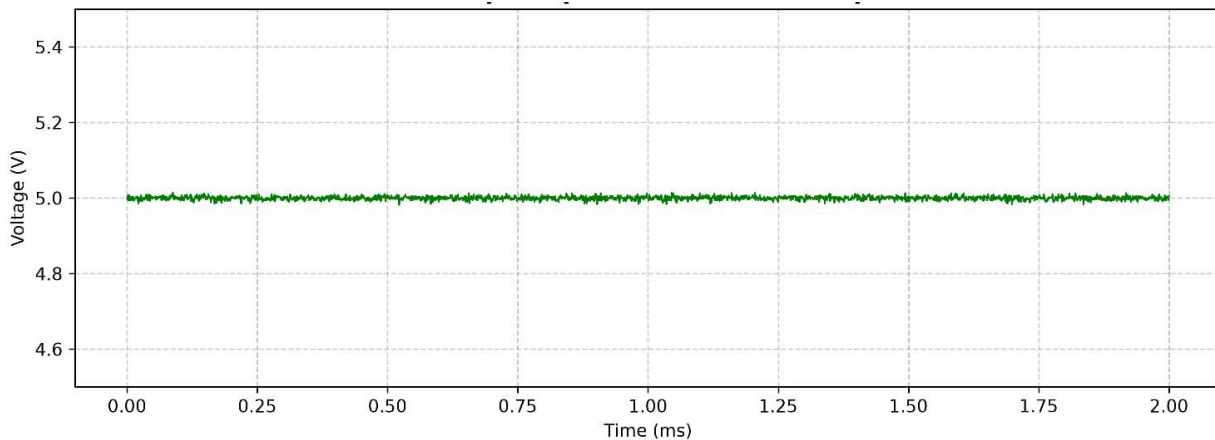
- The capacitor acts like a water tank. When the voltage peaks at 17V, it charges up. When the voltage drops to 0V, it releases stored energy, filling the gap.
- The result is a steady DC voltage with a small ripple (e.g., 15V to 17V).



4. LM2596 Converts to 5V DC:

The LM2596 takes the rippling 15V-17V DC input and chops it at 150kHz

using its internal switching transistor. The inductor and output capacitors then smooth this high-frequency switching back into a perfectly flat 5V DC. Unlike the 7805 (which burns the excess 12V as heat), the LM2596 achieves this with ~85% efficiency, generating only ~2-3 Watts of heat. This allows the charger to safely deliver up to 3A (15W) without thermal shutdown.



Phase 7: 3D Enclosure Design (OpenSCAD)

After successfully testing the circuit on the perfboard, we moved to designing a custom enclosure using OpenSCAD. The goal was to create a compact, functional case that protects the components, manages heat, and allows easy access for inspection and demonstration.

7.1 Design Requirements & Constraints

The design had to accommodate the following physical components and constraints:

- Transformer: 8cm × 5cm × 4.8cm (fixed in place).
- Perfboard with Circuit: 7.3cm × 5cm × 0.4cm (must be removable for testing).

- USB Port: Needs an opening positioned 0.5cm from the short edge and 3cm from the long edge.
- Input Power Wire: Needs a channel for the thick 230V AC wire to exit from the top of the box.

7.2 The Solution: Stepped Deck with Sliding Tray

To meet the accessibility requirement, we designed a sliding tray that sits between the transformer chamber and the perfboard chamber. This tray holds the perfboard and can be pulled out from the side (using a small handle tab) without disconnecting any wires. This allows TAs to inspect the soldered circuit directly.

Key Design Features:

1. Sliding Tray (1.5mm thick)

- The tray has the exact same footprint as the perfboard (7.3cm × 5cm).
- It slides into a slot in the box and can be removed by pulling a small handle tab on the outside.
- This allows the entire PCB to be pulled out for inspection while the transformer and wiring remain fixed inside.

2. Transformer Chamber (Bottom)

- A fixed compartment at the bottom of the box holds the transformer securely.
- The chamber is sized with a 2mm buffer to prevent wobbling during handling.

3. Perfboard Chamber (Top)

- Sits above the sliding tray with a 2.5cm air gap for the LM2596, capacitors, and heatsink.
- Open at the top for easy access.

4. Wire Routing Gap

- Because the transformer (8cm) is longer than the perfboard (7.3cm), a 0.8cm gap remains on one side of the tray. This gap is used to route the thin transformer secondary wires from the transformer up to the perfboard, eliminating the need for extra holes.

5. USB Opening

- Located on the short wall of the box, positioned 0.5cm from the bottom of the perfboard chamber and 3cm from the long edge.
- The opening is 1.7cm × 1.0cm, with a 0.15cm margin added to accommodate print tolerance.

6. Input Wire Channel (Keyhole Design)

- A channel on the back wall allows the thick 230V AC input wire to exit from the top.
- The channel has a rectangular slit (0.6cm wide) that extends from the top of the box down to a "keyhole" shaped opening (a rectangle + semicircle, 0.5cm total height) just below the sliding tray.
- This design allows the wire to be threaded through the slit and rested in the semicircular bottom, then secured by sliding the tray back into place.

7. Friction-Fit Lid

- A separate lid slides over the top of the perfboard chamber.
- The lid has its own USB and wire channel cutouts so they remain accessible when the lid is on.

- It uses a friction-fit skirt (0.2cm wall) with a small clearance (0.15cm) so it can be easily removed for demonstration.

7.3 OpenSCAD Implementation

The entire enclosure was modeled in OpenSCAD using parametric variables. This allowed us to quickly adjust dimensions (e.g., buffer sizes, wall thickness) and regenerate the model. The code is divided into three modules: `main_box()`, `tray()`, and `lid()`, which are rendered together in the assembly view.

Key Variables:

- `wall = 0.2` (2mm wall thickness)
- `fit_buffer = 0.2` (2mm clearance for components)
- `tray_t = 0.15` (1.5mm tray thickness)
- `clear_above_pcb = 2.5` (2.5cm air gap above the PCB)

7.4 3D Printing & Assembly

The parts are printed separately:

- Main Box: Printed with the open top facing up (no supports needed).
- Sliding Tray: Printed flat.
- Lid: Printed flat.

Assembly Steps:

1. Insert the transformer into the bottom chamber.
2. Thread the input wire through the keyhole channel on the back wall.
3. Slide the tray into its slot until flush.
4. Place the perfboard on top of the tray.
5. Connect the transformer wires to the PCB.

6. Slide the lid on.

7.5 Heat Management in the Enclosure

To prevent overheating:

- The transformer chamber and PCB chamber are physically separated by the tray.
- A 2.5cm air gap above the PCB allows heat from the LM2596 to rise and escape.
- The input wire channel and USB opening provide passive airflow.
- The lid is not sealed (friction-fit), so hot air can escape through the gaps.

7.6 Our final code

```
// ----- WHICH PART TO SHOW / EXPORT -----  
PART = "assembly"; // "assembly" | "box" | "tray" | "lid"
```

```
// ===== GIVEN MEASUREMENTS =====  
transformer_l = 8;  transformer_w = 5;  transformer_h = 4.8;  
perfboard_l  = 7.3; perfboard_w  = 5;   perfboard_h  = 0.4;
```

```
usb_w = 1.7;  usb_h = 1.0; // ASSUMPTION: you wrote "1.7*10.8" —  
treated the
```

```
                // trailing ".8" as a typo and kept 1.7 x 1 (matches  
                // every earlier message). Tell me the real number  
                // if this is wrong — it's one variable to fix.
```

```
clear_above_pcb = 2.5;    // air gap above the perfboard's top surface
```

```
tray_t = 0.15;           // 1.5mm, as you specified
```

```
// ===== BUFFERS =====
```

```

wall      = 0.2; // 2mm walls — strong enough to carry the transformer
fit_buffer = 0.2; // static footprint clearance (2mm, "don't overdo it")
near_gap   = 0.1; // small sliding clearance kept on the tray's flush side
vert_clear = 0.1; // total Z clearance for the tray inside its slot
lid_clearance = 0.15; // total clearance for the lid's friction-fit skirt
hole_margin = 0.15; // extra margin on the LID's own USB/wire cutouts,
so
           // print tolerance never blocks them

// ===== DERIVED =====
// interior sized to the BIGGER footprint (the transformer, 8 x 5)
inner_L = max(transformer_l, perfboard_l) + fit_buffer; // 8.2
inner_W = max(transformer_w, perfboard_w) + fit_buffer; // 5.2
outer_L = inner_L + 2*wall; // 8.6
outer_W = inner_W + 2*wall; // 5.6

trans_h = transformer_h + fit_buffer; // 5.0
trans_top = wall + trans_h; // 5.2

slot_h = tray_t + vert_clear; // 0.25
slot_bottom = trans_top - vert_clear/2; // 5.15
slot_top = slot_bottom + slot_h; // 5.40
perf_bottom = slot_top; // floor of perfboard chamber

perf_chamber_h = perfboard_h + clear_above_pcb; // 2.9
outer_H = perf_bottom + perf_chamber_h; // 8.30

// ---- tray (slides along Y, out through the Y=0 wall) ----
gap_total_x = inner_L - perfboard_l; // 0.9 leftover next to
transformer
tray_x0 = wall + near_gap; // 0.3
tray_x1 = tray_x0 + perfboard_l; // 7.6 (far_gap = 0.8
beyond this)

```

```

blade_y = inner_W + wall; // 5.4 reaches flush with
outer front face
handle_w = 3.0; handle_l = 1.5; // grip tab, centered in X

// ---- USB opening (short wall at X = outer_L) ----
usb_y0 = 3.0; // ASSUMPTION: "3cm from the longer
(Y-normal) wall"
usb_z0 = perf_bottom + 0.5; // ASSUMPTION: "0.5cm" read as height
above the tray surface
// -> re-check against your photo before printing (you
flagged
// this hole's position as the one to double-check)

// ---- input-wire channel (back wall, Y = outer_W, inside the 0.8cm gap
column) ----
wire_w = 0.6;
wire_x0 = tray_x1 + (gap_total_x - near_gap - wire_w)/2; // centered in
the gap column
slit_z_top = slot_bottom - 0.5; // "the slit, 0.5cm below the sliding part"
keyhole_rect_h = 0.2;
keyhole_bottom = slit_z_top - keyhole_rect_h - wire_w/2; // rectangle(0.2) +
semicircle radius(0.3) = 0.5 total

echo("outer box L,W,H =", outer_L, outer_W, outer_H);
echo("USB opening Y,Z start =", usb_y0, usb_z0, " size", usb_w, "x",
usb_h);
echo("wire gap column: X", tray_x1, "to", wall+inner_L, " width", inner_L -
perfboard_l - near_gap);

//
=====
=====
// SHARED CUTTERS

```

```

//
=====
=====
module usb_cut(margin=0) {
    translate([outer_L - wall - 0.1, usb_y0 - margin/2, usb_z0 - margin/2])
    cube([wall + 0.2, usb_w + margin, usb_h + margin]);
}

module tray_slot_cut() {
    translate([wall - 0.1, -0.1, slot_bottom])
    cube([inner_L + 0.2, wall + 0.2, slot_h]);
}

// the whole input-wire feature: straight channel from the top down to the
// slit, then a rectangle+semicircle "keyhole" for the last 0.5cm
module wire_channel_cut(margin=0) {
    y0 = inner_W + wall - 0.1;    // through the back (Y = outer_W) wall
    y1 = outer_W + 0.1;
    // upper straight channel
    translate([wire_x0 - margin/2, y0, slit_z_top])
    cube([wire_w + margin, y1 - y0, outer_H - slit_z_top + 0.2]);
    // keyhole rectangle part
    translate([wire_x0 - margin/2, y0, slit_z_top - keyhole_rect_h])
    cube([wire_w + margin, y1 - y0, keyhole_rect_h]);
    // keyhole rounded bottom
    translate([wire_x0 + wire_w/2, (y0+y1)/2, slit_z_top - keyhole_rect_h])
    rotate([90,0,0])
    cylinder(h = y1 - y0, r = wire_w/2 + margin/2, center = true, $fn =
32);
}

//
=====
=====

```

```

=====
// MAIN BOX
//
=====
=====
module main_box() {
    difference() {
        cube([outer_L, outer_W, outer_H]);
        translate([wall, wall, wall])
            cube([inner_L, inner_W, outer_H - wall + 1]); // hollow, open top
        tray_slot_cut();
        usb_cut();
        wire_channel_cut();
    }
}

//
=====
=====
// SLIDING TRAY
//
=====
=====
// Local Y convention: Y=0 is the OUTER tip of the handle (outside the
case);
// Y increases going INTO the box. Blade spans [handle_l,
handle_l+blade_y].
module tray() {
    union() {
        translate([perfboard_l/2 - handle_w/2, 0, 0])
            cube([handle_w, handle_l, tray_t]); // grip tab, outside the
case
        translate([0, handle_l, 0])

```

```

        cube([perfboard_l, blade_y, tray_t]);          // blade: matches
perfboard footprint
    }
}

//
=====
=====
// LID
//
=====
=====
module lid() {
    skirt_depth = outer_H - perf_bottom;
    skirt_wall  = 0.2;
    lid_top_t   = 0.2;
    so_L = outer_L + lid_clearance + 2*skirt_wall;
    so_W = outer_W + lid_clearance + 2*skirt_wall;
    si_L = outer_L + lid_clearance;
    si_W = outer_W + lid_clearance;

    union() {
        translate([-skirt_wall - lid_clearance/2, -skirt_wall - lid_clearance/2, 0])
            cube([so_L, so_W, lid_top_t]);

        translate([0, 0, -skirt_depth])
            difference() {
                translate([-skirt_wall - lid_clearance/2, -skirt_wall -
lid_clearance/2, 0])
                    cube([so_L, so_W, skirt_depth + lid_top_t]);
                translate([-lid_clearance/2, -lid_clearance/2, -0.1])
                    cube([si_L, si_W, skirt_depth + 0.1]);
                translate([0,0,skirt_depth]) usb_cut(hole_margin);
            }
    }
}

```

```

        translate([0,0,skirt_depth]) wire_channel_cut(hole_margin);
    }
}

//
=====
=====
// RENDER
//
=====
=====
if (PART == "assembly") {
    color("SteelBlue")    main_box();
    color("Orange")       translate([tray_x0, -handle_l, slot_bottom]) tray();
    color("LightGray", 0.5) translate([0, 0, outer_H]) lid();
} else if (PART == "box") {
    main_box();
} else if (PART == "tray") {
    tray();
} else if (PART == "lid") {
    translate([0, 0, outer_H - perf_bottom]) lid(); // sits flat, right-side up, for
printing
}

```

How to make our charger better?

1. Upgrade the power source: Upgrade to 12V 3A to eliminate voltage sag and enable full 15W output.
2. Diodes: Replace 1N4007 with Schottky (SR360) to reduce heat and improve efficiency.
3. USB-C PD: Add a trigger module to negotiate 9V/12V for true fast charging (20W+).

4. Thermal management: Install a proper aluminum heatsink for continuous 3A operation.
5. Safety: Add a fuse on the AC input and a TVS diode on the output.